

Shunt Audit: The Road to 9 out of 10

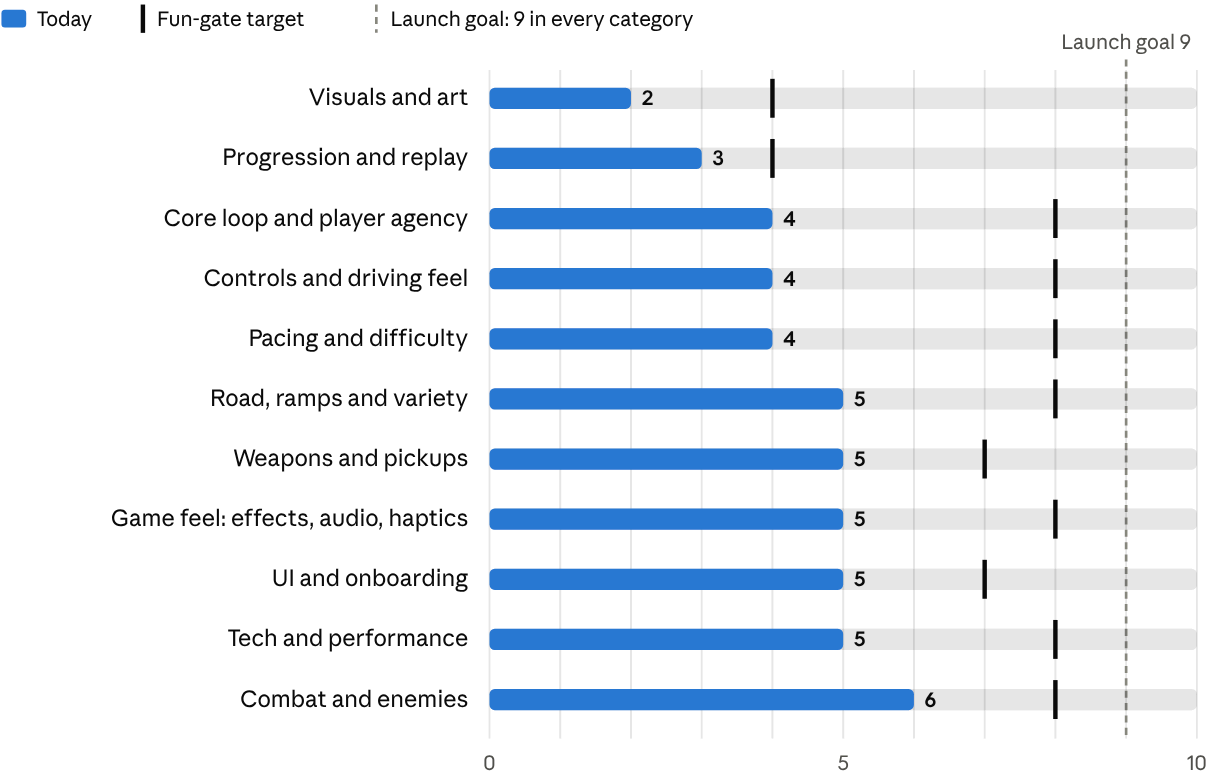
Oct 2, 2026 · @BernoAlpha

Scorecard

This build **scores 4.4 out of 10 today, the average of the 11 category scores in the chart below**. The team shipped nearly every sprint 1 to 4 item from the last review, and the road is finally alive, but three things keep it far from a 9: **the game mostly plays itself, the Slam fires by accident, and it does not drive like a car yet**.

The build averages 4.4 out of 10; most categories must double to reach 9

Reviewer scores, 11 categories · average today 4.4 · average fun-gate target 7.1



reviewer scores · 11 categories · October 2026 audit

Visuals and progression have the widest gaps, but they wait until after the fun gate on purpose. Agency, controls and pacing come next and are the cheapest to fix, so they lead the next two sprints.

What was audited: all 812 lines of the sprint 1 to 4 build. The GitHub Pages link returned 404, so the source is the published "Shunt Graybox" artifact, whose header names the same sprint 1 to 4 build. Ten computer-controlled drivers played it at iPhone 15 size for about 20 minutes of game time: a skilled driver, human-speed thumbs at seven speeds, and an idle driver that never touches the screen after starting.

Measured in the playtest	Result	What it means
Wrecks per minute with zero input	19.6	The game plays itself
Wrecks per minute, skilled driver	22.2	Skill adds only 13% more action
Score in 86 s with zero input	12,355	Points arrive without the player
Accidental Slams at normal steering speed	8 to 14 per minute	The signature move misfires constantly
Armor lost by human-speed drivers in 95 to 190 s	0 to 1 pip	There is no real threat yet
Time with 2 or more cars on screen	98% to 100%	Fixed (last review: no car nearby 47% of the time)
Longest quiet gap, skilled driver	2.6 s	Meets the 3-second rule
Screen width used on iPhone 15	307 of 393 pt, image squashed 22%	One missing CSS line

Credit where it is due: forward-driving traffic, car-to-car collisions with mass, the Bruiser duel, the Gunner van and armored truck, three specials, two gun upgrades, four ramp setups, barrels, forks, lane closures, districts, title, pause and settings screens, a daily run, cash, procedural music and a fixed 120 Hz physics step all landed. That is a lot of good work in one pass. The critique below is about what separates a working prototype from a 9.

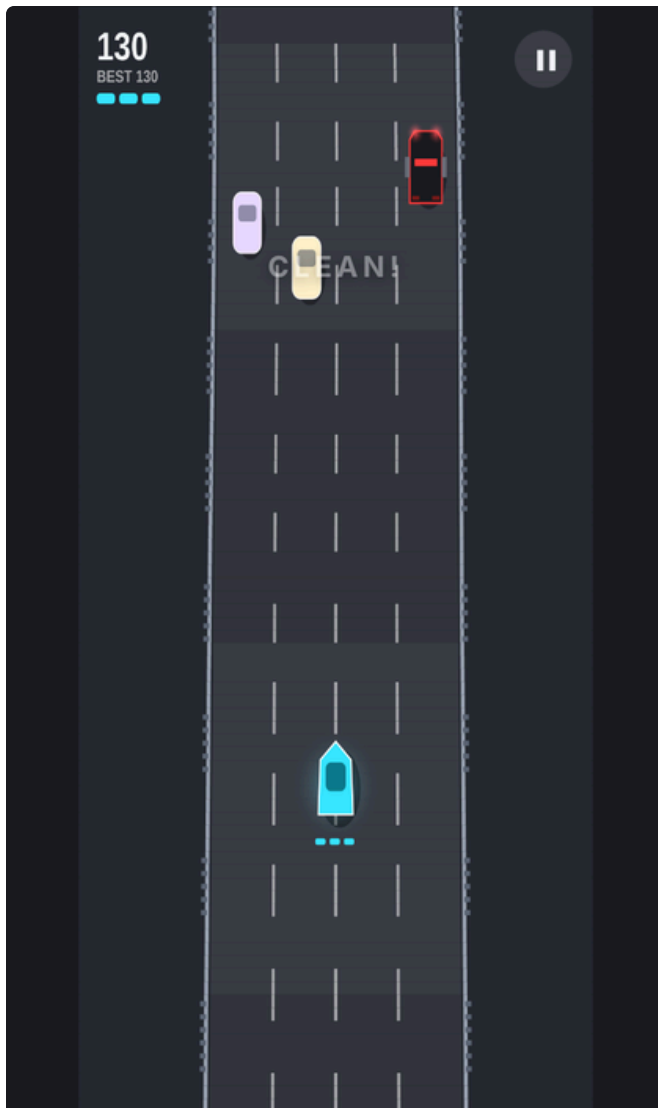
The hard truths

These are the problems that matter most, ranked by how much fun they cost, each with the evidence from the code or the playtest. Fix the first five before adding a single new feature.

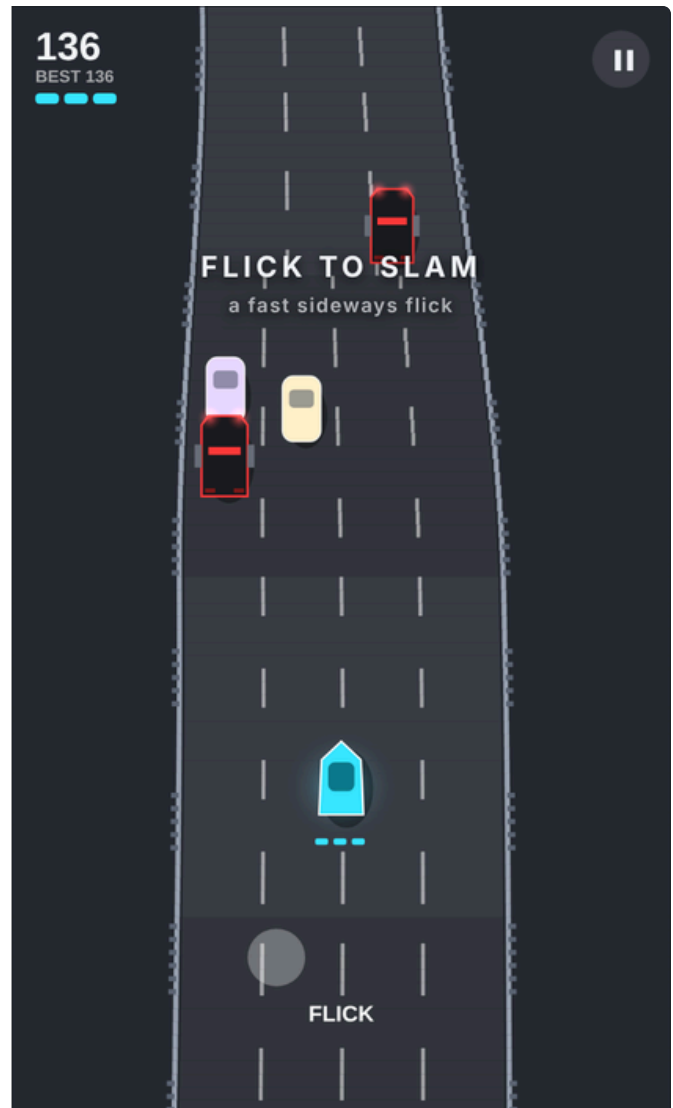
1. **The game plays itself.** A driver that never touched the screen after starting lasted 86 seconds, wrecked 28 cars and scored 12,355. Auto-fire plus the director's habit of spawning a weak car in the player's lane (`script()` and `director()` , the "3-second rule") means kills happen to the player. The best driver wrecked only 13% more cars per minute than the idle one. Players feel this within one run: "I'm not doing anything."
2. **The Slam fires by accident.** The flick test (40 stage points in 120 ms) catches ordinary lane changes. Drivers moving their thumb at 200 to 300 stage pt/s triggered zero accidental Slams; at 450 to 900 pt/s they triggered 8 to 14 per minute, nearly all with no enemy nearby. On an iPhone 15 the threshold works out to roughly 200 screen points per second, a gentle swipe. Each misfire throws the car sideways at 950 pt/s: from the center lane it slammed into the rail 107 pt away in 0.15 s. Those burst numbers came from my last review and they were too strong; the Driving model section corrects them.
3. **The picture is squashed and letterboxed.** `#stage` is a flex item, so it shrinks to the screen width before `fit()` scales it. On an iPhone 15 the 503-wide canvas is displayed 393 wide and then scaled down again: the game uses 307 of 393 points and every shape is 22% too narrow (33% on an iPhone SE). The fix is one line: `#stage { flex-shrink: 0; }`.
4. **There is no threat.** Human-speed drivers lost 0 or 1 armor pip in 95 to 190 seconds; the skilled driver lost one pip in almost 4 minutes and got it back from a crate. Enemies telegraph clearly (good) but almost never land a hit, and armor crates refill the bar. Without danger, wrecks feel cheap.
5. **It does not drive like a car.** Sideways speed is capped at 520 pt/s while forward speed starts at 420 pt/s, so the car strafes faster than it drives, with only a 10 degree lean. It moves like a cursor or a hovercraft. There is no throttle, no brake, no drift, and forward speed is just a timer (420 pt/s rising to 700 pt/s over 5 minutes).
6. **The signature move pays nothing.** In the skilled run, Slam kills earned 200 of 55,950 points (0.4%), while 18 chain wrecks earned 24,200 (43%). The reward structure tells players to ignore the Slam.

7. **There is nothing to steer around.** The road's center wanders up to 70 pt every 600 pt of road. There are no corners, no hills and no reason to brake, so drifting has nowhere to happen.
8. **Speed does not read.** The shoulders are flat color with no trees, posts, signs or buildings passing by, which are the main cue for speed in a top-down game. The lane dashes, the only moving texture, are drawn in 10-pixel road slices, so they advance in 10-pixel steps instead of gliding.
9. **Feedback is missing on the phone.** Haptics still use `navigator.vibrate`, which does nothing on iPhone. The death replay is recorded 30 times a second (`G.replay`) and never shown, and `G.bestMoment` is set and never used.
10. **Screen shake moves the whole canvas.** `kickShake` translates the canvas element itself, so a big hit uncovers a black strip of up to about 18 pt at the screen edge. Shake the camera inside the renderer instead, with overscan.
11. **Small correctness issues.** The gun runs on real time outside the fixed physics step, so it keeps firing at full rate during slow motion. Road lookups scan every road knot from the start (`Road.at()`), which costs 4.6 times more per call after 10 minutes (0.26 to 1.17 microseconds, about 7,200 calls a second). Cheap to fix now, annoying to find later.

The squashed screen from hard truth 3, at iPhone 15 size, as shipped and after the one-line fix:



As shipped: 307 of 393 pt used, every shape 22% too narrow



After the one-line fix: full width, true proportions

Driving model: throttle, brake, slide and drift

Replace the "car follows the thumb sideways" model with a real arcade car: the thumb still says where you want to be, but the car gets there by **turning its nose, gripping, sliding and drifting**, and the player controls speed with brake, drift and boost. This is the single biggest change on the road to a 9.

Controls (two thumbs, still playable with one)

Input	Action	Why
Left thumb, slide anywhere on the left 60%	Steer (relative drag, as today)	Keeps the precise "car goes where I point" feel
Right thumb, hold the pedal pad (bottom right)	Brake; hold while steering = drift	One pad gives slow down, slide and drift
Release the pad after a drift	Mini-turbo boost	The reward that makes drifting addictive
Special button (above the pedal pad)	Fire the special	Same as today, moved up
Flick toward an enemy that is beside you	Slam (contextual, see below)	Fixes the accidental Slams
Settings: "Auto-drift" on	Hard steering at speed drifts by itself; the pedal pad becomes brake only	One-thumb players and younger players

Speed: the player is in charge of it

Parameter	Target value
Cruise speed (auto throttle, no input)	480 pt/s, rising 10% per district
Top speed on the flat, with boosts	900 pt/s
Nitro	1,150 pt/s for 1.5 to 2 s
Downhill drops (see below)	1,400 pt/s and up
Brake deceleration	900 pt/s ² down to a 260 pt/s minimum
Throttle back up after braking	0 to cruise in 1.2 s
Slipstream: 0.8 s directly behind any car	+120 pt/s boost for 1 s, with a "draft" whoosh

Forward speed should never again be just a clock. Speed comes from drafting, drift boosts, clean landings, nitro and hills, and you lose it by braking, scraping rails and taking hits. Going fast becomes a skill and a reward.

Heading-based steering (makes it look and feel like a car)

- The thumb sets a target lane position, as now. The car reaches it by **yawing**: up to 30 degrees of heading in grip, at up to 240 degrees per second of turn rate.
- Sideways speed comes from heading: lateral = speed $\times$ sin(heading). At 480 pt/s and 30 degrees that is 240 pt/s; at 900 pt/s it is 450 pt/s. **Faster means more agile**, which rewards speed. Today's flat 520 pt/s strafe at 420 pt/s forward has to go.
- The car body always points the way it is actually moving, plus the slip angle when sliding. No more 10-degree cosmetic lean on a car moving sideways at 50 degrees.
- Tire grip limit: 1,500 pt/s² of sideways acceleration. Ask for more (a hard turn at speed, a corner, a hit) and the rear lets go into a slide.

Drift (the Tokyo drift moment)

- Hold the pedal pad while steering: rear grip drops to 35%, the tail swings out to a **35 to 55 degree slip angle**, and the player holds the angle with small counter-steering movements of the thumb.
- The car loses 5% of its speed per second while drifting, so a drift is a choice, and corners (see Roads with real turns) make it the fastest line.
- **Drift charge** builds with time $\times$ slip angle: tier 1 at 0.8 s, tier 2 at 1.6 s, tier 3 at 2.6 s, shown by spark color from the rear wheels. Release for a mini-turbo of +150, +250 or +400 pt/s for 0.6, 0.9 or 1.2 s.
- **Drift points** bank on a clean exit: slip angle $\times$ speed $\times$ time, multiplied by the combo. Hit anything mid-drift and the bank is lost.
- **Drift Slam**: an enemy beside you when your tail swings out takes a tail hit at Slam power. The drift becomes an attack, which is the spy car fantasy.
- Exit without fishtailing: grip returns over 0.25 s, with one light wobble that the player can catch.

The Slam, corrected

- My last review specified a 950 pt/s burst for 0.22 s. That covers 209 pt, more than three lanes, which is why misfires throw the car into the rail. **Use 600 pt/s for 0.11 s: about 66 pt, one lane.**
- Make it **contextual**: a flick fires a Slam only if an enemy is within 1.4 lanes to that side and within 60 pt ahead or behind. With no enemy there, the flick is just a fast lane change.
- Detect flicks properly: at least 60 stage points in under 100 ms, starting from a thumb that moved less than 10 pt in the previous 120 ms, using each touch event's `timeStamp` and coalesced events rather than the frame clock.

- Target after these changes: **under 1 accidental Slam per 10 minutes** at normal steering speeds.

Smooth movement

The game runs at a steady 60 fps, but motion is not smooth yet: the main moving texture stutters, physics states are drawn without interpolation, and the code allocates thousands of throwaway arrays per second. Smoothness is a feature; players read it as quality before they read anything else.

What the playtest measured

- **Lane dashes judder.** The world advanced a steady 7 pt every frame, but a lane dash on screen moved 9, 8, -1, 9, -1, 9 and 0 pixels on successive frames. The road is painted in 10-pixel horizontal slices (`drawRoad` , `step = 10`), and each slice is either all dash or no dash, so the dashes jump instead of gliding. Rails stair-step on curves for the same reason.
- **No interpolation.** Physics runs at a fixed 120 Hz (good), but the renderer draws whatever the last step left. When a frame lands between steps, or runs 1 step one frame and 3 the next, objects jitter by a few points.
- **Garbage every step.** The physics step creates about 21 new arrays (`filter` calls) plus new objects for sparks, marks and debris, 120 times a second, about 2,500 throwaway arrays a second, and the unused replay buffer snapshots every nearby car 30 times a second. On iPhone this shows up as periodic hitches when the garbage collector runs.
- **Canvas shake.** Shaking the canvas element with CSS moves the whole picture and uncovers black edges (see the third screenshot in the UI section).

Requirements for the build team

1. Draw the road as continuous shapes (polygons per segment, or a GPU mesh), never fixed-height slices. Lane dashes, rail posts and road texture must move by exactly the distance traveled each frame.
2. Interpolate rendering between the last two physics states (`alpha = accumulator / step`). Cars, camera and particles all draw at the interpolated position.
3. Zero allocations in the physics step: pre-allocated pools for cars, bullets, particles and decals; compact arrays in place instead of `filter` .
4. Camera on critically damped springs (no overshoot) for follow, look-ahead and zoom. Shake is an offset added inside the renderer, with 24 pt of overscan so edges are never exposed.

5. Read touches with each event's `timeStamp` and `getCoalescedEvents()`, apply them at the start of the next physics step, and keep the thumb-to-car response under 2 frames.
6. Support 120 Hz ProMotion: render at the display rate, keep physics at 120 Hz.

Frame budget on the minimum device (iPhone 12): 99% of frames under 16.7 ms at 60 Hz, or under 8.3 ms at 120 Hz on Pro models, measured over a 10-minute session with no hitch longer than 33 ms.

Tire smoke, skid marks and every driving effect

Today there is **no tire smoke at all**: skid marks are pairs of tiny rectangles dropped when the car moves sideways fast, and the only smoke is five gray circles behind a damaged car. Drifting only feels like drifting when the screen fills with smoke, rubber and sparks, so this list is part of the driving model, not polish.

Effects spec (every row ships before the downhill sprint)

Effect	Trigger	Spec	Today
Tire smoke	Rear-wheel slip over 8 degrees, burnouts, hard braking	Per rear wheel: 0 puffs/s in grip, 40/s at 20 degrees of slip, 90/s at 45 degrees. Each puff starts at 8 pt and grows to 48 pt over 1.4 s, fading from 55% opacity, keeps 30% of the car's speed, then drifts and curls	Missing
Skid marks	Slip over 8 degrees or locked brakes	Continuous ribbon under each rear wheel, 6 pt wide, darker with more slip, fades over 6 s; drift and brake marks look different	Dotted rectangles
Drift sparks	Drift charge tiers	Spark stream from the rear wheels whose color shows the tier; a burst on release	Missing

Effect	Trigger	Spec	Today
Burnout	Title screen and every restart	Car spins its rear wheels in a cloud of smoke before launch	Missing
Brake lights	Braking	Red glow on the road behind the car, plus brake dust	Missing
Exhaust and backfire	Lifting off the throttle, boosts	Pops and flame licks; nitro adds two flame cones, heat shimmer and speed lines	Speed lines only
Rail sparks	Scraping a rail or wall	Continuous spark stream along the contact side, plus a metal screech	1 spark per step
Slam impact	Slam or drift slam contact	Spark fan along the contact side, a crumpled-panel decal, glass glints	Sparks, crunch
Wreck	Any enemy wreck	Fireball, shockwave ring, 3 to 6 tumbling debris pieces (doors, hubcaps, a bouncing tire), a burning hulk for 3 s and a scorch mark	Glow and dots
Landing	End of a jump	Dust ring, suspension squash, sparks on a hard landing	Ring, squash
Surface spray	Driving on the shoulder, district surfaces	Dust in the desert, puddle spray in the city, snow spray on the mountain	Missing
Weather	District mood	Rain streaks and road spray behind every car; heat haze in the desert	Missing
Lights	Night districts	Headlight cones on the road, enemy red light bars, neon reflections on wet asphalt	Red headlight glows

Effect	Trigger	Spec	Today
Smoke color	Cosmetic unlock	Colored tire smoke, a signature reward players love to show off	Missing

Rules that keep effects from ruining the game

- **Effects everywhere, but never over a threat.** Smoke and debris draw below enemies, or enemies get a red outline whenever smoke covers them. Nothing may hide an enemy's tell for longer than 0.3 s.
- **Budget:** 600 live particles on an iPhone 12, with a lower setting for older phones and Low Power Mode. Pool everything (see Smooth movement).
- **Pre-rendered sprites only:** soft puffs, sparks and glows drawn once into textures, never as gradients created every frame (the current code has fixed this for glows; extend it to everything).
- Every effect has an audio partner: tire squeal that rises with slip angle, a rubber chirp on grip recovery, a screech on rails, a whoosh on near misses.

Roads with real turns

Drifting needs corners, and today there are none: the road's center only wanders up to 70 pt every 600 pt, so the player never brakes, never chooses a line and never slides. Add real corners, and rotate the top-down camera with the road so "forward" always points up the screen.

How to build it without a rewrite

- Keep simulating in road space, as the code already does: distance along the road plus lateral offset. Add **curvature** to each road segment.
- Bending happens in two places only: the renderer (each segment is rotated and translated along the road's heading) and physics (a sideways push of $\text{speed}^2 \times \text{curvature}$ toward the outside of the corner).
- Build corners from constant-curvature arcs joined by smooth transitions, so the steering load builds and releases gradually instead of snapping.
- The camera rotates with the road heading on a spring, leading about 0.4 s into the corner so the player can see the exit.

Corner types (with the grip limit of 1,500 pt/s² from the Driving model section)

Corner	Radius	Fastest speed in grip	What the player does
Sweeper	1,200 to 2,000 pt	Flat out at any speed	Holds the racing line, fights at speed
Fast bend	500 to 700 pt	870 to 1,020 pt/s	Lifts slightly or drifts at top speed
Hard corner	about 300 pt	670 pt/s	Brakes or drifts; drift carries more speed
Hairpin	about 180 pt	520 pt/s	Must brake or drift; the big drift-points moment

Road rhythm

- Alternate **combat straights** (30 to 45 s, 4 to 5 lanes, gentle sweepers, heavy traffic and enemies) with **technical sectors** (15 to 25 s, 2 to 3 lanes, 3 to 6 corners, light traffic, big drift points).
- Traffic slows for corners, so drifting past it is satisfying. Enemies shunted in a corner slide toward the outside wall, which makes corners the best place for Slams.
- Warn every corner 2 s ahead: chevron boards on the outside, a painted arrow on the road and a color code (yellow: lift, red: brake or drift).
- Corner furniture: rumble strips on the inside (buzz and a light haptic), tire walls and barriers on the outside (wreck enemies into them), cones and spectators at hairpins.
- Hill crests on straights: the car goes light over a crest and gets small air, a taste of the downhill drops.

Downhill drops: the SSX Tricky moment

Every 90 to 120 seconds the highway tips over a mountain edge and the camera swings down behind the car for a 30 to 45 second **drop**: huge speed, big air, tricks that turn into boost, and rivals to knock off the road. This is the moment players will record and share, and it gives the game a second gear that most car brawlers do not have.

What makes SSX Tricky's downhill so fun, and the car version of each

In SSX Tricky	In this game
Speed that keeps building on the slope	Gravity takes the car from 900 to 1,400+ pt/s; wind roar, a wider camera field of view and speed lines sell it
Constant kickers and big air	A kicker every 3 to 5 s: 1.5 to 4 s of air, and one huge jump per drop with 5 s of air and a slow-motion beat
Tricks fill a boost meter, and a full meter unlocks bigger tricks	Car tricks fill a Heat meter ; a full meter triggers Overdrive (8 s of unlimited nitro, double points, signature stunts worth 5x)
Risk on every landing	Land wheels-down within 25 degrees for boost; 25 to 50 degrees costs speed; worse is a tumble and an armor pip
Rivals you can knock over	3 to 5 rival cars come down with you: bump them off the edge, shoot them, or land on them
Multiple routes and shortcuts	Each drop forks into a fast safe road, a risky shortcut with bigger jumps, and a hidden route with collectibles

Air tricks (simple gestures, any thumb, only in the air)

- Swipe up: front flip. Swipe down: backflip. Swipe left or right: barrel roll. Draw a circle: flat 360 spin. Hold both thumbs: a stunt pose (the driver leans out and fires).
- Points scale with rotation and air time. A trick must finish before the wheels touch, or the landing counts as a crash.
- Chain tricks in one jump for a multiplier, and keep the combo alive across jumps by landing clean.

Camera: one main view, one special view

- The top-down camera stays the main perspective for combat. The chase camera is used **only** during drops.
- The switch is a 1.2 s eased blend at a clear gate ("DROP AHEAD" sign, the road tilts down). Target chase framing: about 15 degrees of pitch, 9 m behind and 2.5 m above the car, field of view 70 widening to 90 degrees with speed.

- In big air the camera pulls back and lowers to show the height; on landing it punches in briefly.
- Never switch cameras mid-fight. Return to top-down through a tunnel or a canyon exit so the change feels natural.
- Controls do not change between cameras: the thumb still steers by relative drag.
- Audio: in big air, muffle the music and leave the wind; slam it back to full on landing.

What this means for the tech: decide now

- A camera behind the car, hills and a horizon need a real 3D world. The current 2D canvas with its fake perspective cannot do a convincing chase camera.
- Move the prototype to a 3D engine now, before art starts: three.js for the web prototype, then, at the fun gate, either a native shell around the three.js game or a Unity port (see Performance and iOS). The top-down view becomes a high 3D camera and the drop is a low one, in the same world.
- Prototype the drop as its own small level first (one kicker line, one rival, the camera blend, three tricks) and only fold it into the run once it is fun alone.

Combat and enemies

The combat systems work (mass, shunts, chain wrecks, telegraphs), but the **player is not the cause of most of the action and is almost never in danger**. Fix who gets credit and who gets hurt before adding enemies.

1. Give the kills back to the player

- Stop spawning fodder in the player's lane. The "3-second rule" filler (`script()` and `director()`) drops a weak car directly in front of the gun, which is why a driver with zero input wrecked 19.6 cars a minute. Spawn fodder one or two lanes over, so lining up a shot is a decision.
- Score by cause: a wreck the player caused (Slam, drift slam, shunt into something, stomp, missile, oil, a barrel the player shot) pays full points; enemy-on-enemy accidents pay nothing. Today 18 chain wrecks earned 43% of the skilled run's score, and the idle driver got 5 chain wrecks with zero input.
- Make the car the main weapon: Slam, drift-slam and shunt kills pay 3x a gun kill. The twin guns soften targets and finish weak cars.
- Remove the "+0" score pop that appears on every Slam contact (`addScore(T.score.slamKill * 0, ...)` in `contactPlayer`).

- **Targets:** an idle driver wrecks fewer than 5 cars a minute; a skilled driver wrecks 3 times as many as an idle one.

2. Make enemies dangerous

- Damage is halved for the first 60 s, a pip comes back at 30 s, and crates drop armor a third of the time. Keep the halving for 30 s only, drop the free pip, and make armor rare (supply truck only).
- Enemies should attack in combinations: a Bruiser holding beside you while a Gunner lines up behind, two Bruisers boxing you in, a bike pack swarming the lane you need.
- Add the **PIT move**: an Interceptor taps your rear corner and spins you unless you counter-steer, which ties combat to the drift model.
- **Targets:** an idle driver dies within 30 s once past minute 2; a good player's first death comes between minutes 4 and 8; a top player can last 15 minutes.

3. Visible damage on enemies: smoke at half health, sparks and a dragging bumper near death, so players see their hits working.

4. Enemy roster (4 today, 14 at launch)

Enemy	Status	Behavior	Best answer
Weak car	Built	Gun fodder	Line up and shoot
Bruiser	Built	Holds beside you, telegraphs, swerves	Slam first, during hold or tell
Gunner van	Built	Sits behind, red sight line for 0.8 s, fires	Change lanes, then oil slick it
Armored truck	Built	Two lanes wide, immune to guns	Missiles, or shunt something into it
Blade car	New	Extends wheel blades and sideswipes	Out-brake it, or drift slam its tail
Interceptor	New	Fast chaser that attempts the PIT move	Counter-steer, then Slam it into a wall
Bike pack	New	3 to 4 bikes swarm your lane	One Slam scatters the pack into a chain
Mine layer	New	Drops mines from ahead	Shoot the mines for chain points

Enemy	Status	Behavior	Best answer
Tanker	New	Explodes in a 120 pt fireball when wrecked	Wreck it next to other enemies
Hacker van	New	EMP that disables your special for 3 s	Kill it first; it glows before firing
Drone swarm	New	Small, shootable, drops bombs on your lane	Spread shot, or dodge the shadows
Helicopter	New	Strafes a lane after a searchlight sweep	Missiles, or stay out of the light
Rival racer	New, drops only	Races you downhill and bumps you off kickers	Bump it first or land on it
Boss: armored convoy	New, every 3 min	3 armored trucks and escorts, two attack patterns	Missiles on the escorts, Slams into the trucks

5. Civilians

- Keep the -100 penalty (unmultiplied) and the Close Call bonus. Give civilians panic behavior: swerve, brake hard, honk, so weaving through them feels alive.
- In corners, civilians are the obstacle that makes the racing line interesting.

The arsenal: weapons, gadgets, car accessories and combos

The build has 3 guns and 3 specials, all found mid-run and lost at death. A 9 out of 10 version has **a loadout the player builds between runs, upgrades found during the run, and weapons that combine**, so every run plays differently and players experiment.

Loadout: six slots on the car

Slot	Holds	How it is used
Front	Primary gun or ram	Automatic
Rear	Drops and countermeasures	Special button
Roof	Launchers, drones, turrets	Special button (second press, or swap)

Slot	Holds	How it is used
Wheels and chassis	Blades, hop, tires, armor	Passive or gesture
Engine	Nitro, turbo, slipstream	Pedal pad and pickups
Gadget	One spy gadget per run	Double-tap the special button

Weapons and gadgets (26 total; the 6 marked Built exist today)

Item	Slot	What it does	How it changes play
Twin machine guns	Front	Fast straight shots	Built. Baseline
Spread shot	Front	3-way cone	Built. Clears wide roads
Heavy cannon	Front	Slow shells with knockback	Built. Shoot enemies into walls
Laser lance	Front	Beam pierces every car in the lane	Rewards lining enemies up
Flamethrower	Front	Short cone; burning cars panic and swerve	Close-range brawling
Harpoon	Front	Cable into a car ahead: yank it into a wall, or slingshot past it	Hook a pole in a corner for a grapple drift
Ram plow	Front	Slams cost no armor, +50% shunt power	Pure demolition build
Shock ram	Front	Slam stuns the target and arcs to cars next to it	Turns one Slam into a chain
Oil slick	Rear	Cars that cross it spin out	Built. Answer to chasers
Smoke screen	Rear	Pursuers lose you and collide	Escape tool, chain setup
Spike strip	Rear	Pops pursuers' tires	Deadly before corners
Proximity mines	Rear	Explode when a car passes	Area control
Flame trail	Rear	Rear burner leaves a fire line	Combines with oil (see combos)

Item	Slot	What it does	How it changes play
Homing missiles	Roof	Lock on and arc to targets	Built. Kills armor
Rocket pod	Roof	Salvo of 6 straight rockets	Burst damage, skill aim
Attack drone	Roof	Orbits and shoots for 15 s	A wingman
Rear turret	Roof	Auto-fires backward	Covers your blind spot
EMP pulse	Roof	Stalls every enemy on screen for 2 s	Panic button
Wheel blades	Wheels	Sideswipes shred enemy tires	Rewards side contact
Hydraulic hop	Wheels	Jump on demand, about 0.6 s of air	Hop over traffic, stomp on demand
Drift tires	Wheels	Longer drifts, faster drift charge	Drift build
Nitro	Engine	Short surge, invulnerable while it lasts	Built. Smash through jams
Turbo	Engine	Bigger mini-turbo after drifts	Drift build
Holo decoy	Gadget	Enemies chase a hologram for 4 s	Sets up ambushes
Shield	Gadget	Blocks all damage for 3 s	Saves a run once
Glider wings	Gadget	Extends air time in downhill drops	More tricks, longer jumps

Combos (discovering these is half the fun)

Combo	Result
Oil slick + flame trail or flamethrower	A wall of fire across the road
EMP + Slam	Stunned cars fly twice as far
Smoke screen + mines	Blind pursuers drive into the mines
Harpoon + corner	Slingshot around the corner at full speed
Hydraulic hop + explosive barrel	Stomp the barrel, chain everything nearby
Nitro + ram plow	An unstoppable battering ram through traffic

Combo	Result
Spike strip + corner	Pursuers spin into the outside wall
Missiles + tanker	The biggest chain wreck in the game

How weapons arrive during a run

- The supply truck offers a **choice**: two crates on its ramp, one per lane. Drive into the one you want. A choice is far more engaging than a fixed rotation (today the truck cycles missiles, oil, nitro).
- Picking the same item again levels it up to level 3, with a visible change at each level (bigger slick, double missiles, longer nitro).
- Crates drop from wrecks the player caused, never from passive wrecks.

Car accessories

- **Functional mods with trade-offs**: light body (faster, less armor), armored frame (more armor, slower acceleration), sport suspension (better landings), long-range tank (more nitro).
- **Cosmetics**: paint and wraps, decals, underglow color, tire smoke color, exhaust flame color, rims, horn sound, license plate text, hood ornament, and a victory burnout animation.
- **Cars (garage)**: Interceptor (balanced), Muscle (heavy rammer), Rally (drift king), Supercar (top speed), Armored van (tank), each with its own stats and handling.

Game feel: camera, impacts, audio and haptics

The foundations are in (scaled hit-stop, a spring-based shake, speed zoom, a stomp punch-in, 18 procedural sounds and 3-layer music), so the work now is **speed, weight and the new driving moments**: drift, corners, the drop, big air.

Camera

- **Show more road at speed**. At 900 pt/s the player needs about 1.6 s of road ahead. Zoom out up to 20% with speed and slide the car lower on screen; today zoom changes only 5%.
- **Lead into corners and drifts**: rotate with the road heading, and roll about 5 degrees into a drift while pulling out slightly.

- **Fix the shake feel.** The current spring (stiffness 180, damping 14) is underdamped, so every kick wobbles. Use a trauma-based shake: noise-driven offset and up to 2 degrees of rotation, scaled by trauma squared, decaying smoothly. Cap it at 6 pt for a gun kill, 10 for a wreck and 16 for a stomp.
- **Moments get a beat:** 0.25 s of slow motion and a 4% push-in for chain wrecks of 3 or more, boss kills and Overdrive starting.

The new moments, fully specified

Moment	Camera	Sound	Haptic (native)
Drift	Roll 5 degrees into the drift, pull out 5%	Tire squeal rising with slip angle, gravel crunch on the shoulder	Continuous texture whose strength follows slip angle
Mini-turbo	Push-in 3%, speed lines	Turbo whoosh pitched by tier	Sharp tick, then a short rumble
Corner kerb	Light vertical judder	Rumble-strip buzz	Rapid light taps
Slipstream	Slight zoom out	Air hiss rising, then a "pop" when the boost fires	Light tick
Drop start	1.2 s blend to the chase camera	Music filter opens, wind rises	Long soft swell
Big air	Pull back and lower, 0.3 s slow motion on the biggest jumps	Music muffled, wind only	None (silence sells height)
Clean landing	4% punch-in	Bass thud, music slams back in	Heavy transient
Crash landing	Tumble camera for 0.8 s	Metal, glass, scrape	Long heavy pattern
Slam / drift slam	Directional kick away from the hit	Crunch with bass and glass	Sharp heavy transient
Chain wreck x3+	Slow motion 0.25 s, 4% push-in	Rising stinger per chain step	Pulse per step

Audio for production (the graybox synth is fine for now, but plan this)

- Engine in three recorded layers (idle, mid, high) crossfaded by speed, plus gear shifts and backfires on lift.
- Doppler whooshes on every car you pass close to; wind noise that rises with speed.
- Impacts built from layers: metal crunch, glass, sub-bass. A mix limiter so 10 explosions at once never distort.
- Original adaptive music in stems (surf-rock guitar over breakbeats fits a spy car brawler), with a separate drop track and stingers for combo tiers. No licensed spy themes.

Haptics: `navigator.vibrate` does nothing on iPhone, so today every buzz in the code is silent. The native build uses Core Haptics with the patterns above, a continuous drift texture, and a settings toggle (already built in the settings screen).

Art direction: the theme, the landscape and when art starts

Start art direction now, but do not put final art in the game until the plain-shapes version passes its fun gate. Deciding the look early is cheap and steers the engine and camera choices; building final art early is expensive and locks in mechanics that are still changing.

Three directions to choose from

Direction	Look	Strengths	Risks
A. Neon Noir	Night cities, wet asphalt, neon reflections, smoke lit by signs	Perfect for Tokyo-style drifting; dramatic tire smoke; high contrast	Crowded synthwave look; a dark palette can hurt readability
B. Retro Spy Postcard	1960s spy-film pop: bold flat colors, sunny coast roads, alpine passes, mid-century graphic design	Distinctive and timeless; fits the spy fantasy; snowy passes suit the downhill drops	Needs strong graphic design to avoid looking flat
C. Toy Diorama	Miniature world, chunky low-poly cars, tilt-shift blur	Very readable from above; cheap to build; charming	Less "cool"; weaker fit for spy action

Recommendation: B as the base, with A for the night districts. A world tour of spy locations, each with its own time of day and palette, in one stylized low-poly style with flat shading, crisp rim lighting and 1960s spy typography in the UI:

District	Time of day	Signature
Riviera Coast	Late morning	Cliff roads above the sea, tunnels, villas
Desert Canyon	Sunset	Long straights, dust plumes, a dam crossing
Neon City	Night	Tokyo-inspired neon streets, wet roads, hairpins between towers: the drift district
Alpine Pass	Snowy afternoon	Hairpins, then the downhill drops with snow spray and huge kickers

The landscape is a gameplay tool, not decoration

- **Near layer** (rails, lamp posts, signs, trees, every 20 to 60 pt): the main speed cue in a top-down view. Today the shoulders are flat color, which is a big reason the car feels slow.
- **Middle layer** (buildings, cliffs, billboards, parked cars): sets the place and frames the road.
- **Far layer** (sea, fields, rooftops, mountains in the drops): mood and scale, with parallax.
- Vertical objects cast shadows across the road, so motion reads even in a glance.

Rules every piece of art must pass

- Role colors survive in every district: player cyan, enemies black with red, civilians pastel, pickups gold, hazards orange. Test each district with a blur and color-blind filter.
- Silhouette first: an enemy is identifiable from its shape alone at 25% size.
- Effects never cover threats (see the effects rules above).
- Performance budget: texture atlases, baked lighting, emissive neon instead of real-time lights, a fixed draw-call budget per district.

When art starts

1. **Now, in parallel with Sprint A (no integration):** a mood board, three style frames (one per direction), a pick, and a one-page art bible: palettes per district, role colors, silhouettes, scale.

2. **After the driving and combat gates:** a *readable blockout*: final car proportions, final camera height, role colors, primitive roadside props for speed cues. Art that serves gameplay.
3. **After the fun gate:** a *vertical slice*: one district (Neon City) and one downhill drop at final quality, with effects and audio, then a playtest to confirm art does not hurt readability or frame rate.
4. **Production:** the other districts, the garage cars, weapon visuals, UI art, app icon and store screenshots.
5. **Polish:** lighting, post-processing, level-of-detail and performance passes.

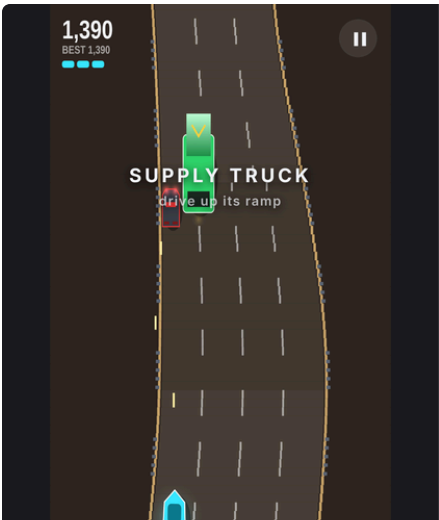
The roadmap drawing at the end shows where these land against the gameplay sprints.

UI, onboarding and screens

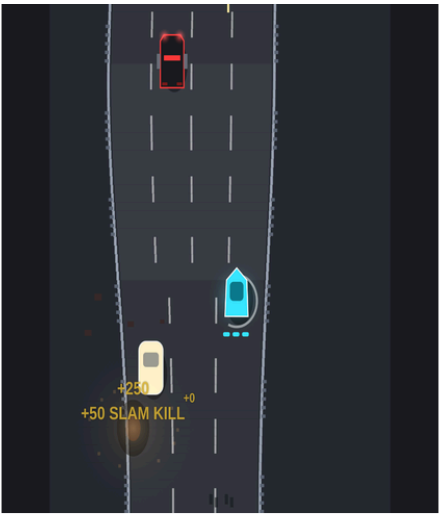
The screens exist now (title, pause, settings, game over, a 3-2-1 resume), which is real progress, but text keeps landing on top of the action and the settings screen looks unfinished.

Problems seen in the screenshots

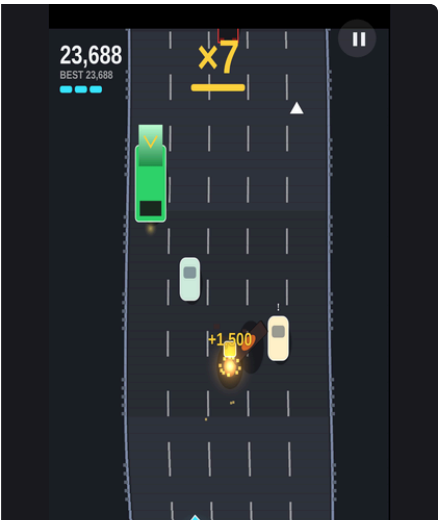
- **Callouts cover what they describe.** Callouts sit at 22% of the screen height, right where new cars arrive: "SUPPLY TRUCK" was drawn on top of the supply truck, and "FLICK TO SLAM" over the Bruisers it was teaching. Move callouts into a slim band under the HUD, or attach short labels to the object itself with a pointer.
- **The top of the screen is the most dangerous place.** Threats enter from the top, and the combo counter (up to 58 pt tall at x8) sits in the top center. Keep the HUD in the corners: score top left, armor under it, combo next to the score.
- **Settings looks unfinished.** Labels are tiny, crowded to the left edge and wrap onto two lines ("SCREEN / SHAKE"), while the buttons are oversized white pills. Use standard rows: a 17 pt label on the left and a switch on the right, grouped into Audio, Controls and Accessibility.
- **Progress leads nowhere.** Game over shows "+cash, next car at 5,000" with a progress bar, but there is no car to buy. Do not promise a reward the game cannot give yet.
- **The death replay is recorded but never shown,** so players still do not see what killed them.



"SUPPLY TRUCK" drawn on top of the truck



A "+0" pop on every Slam, next to the real score



Shake uncovers a black strip; the x7 combo covers incoming cars

Onboarding for the new controls (teach by doing, one thing at a time)

Time	Teach	How
0 to 3 s	Launch	Burnout, then a jump off the garage ramp. No text
3 to 10 s	Steer	Ghost thumb on the left pad; a lane of cones to weave through
10 to 20 s	Slam	A slow Bruiser holds beside you; time slows until the player flicks toward it
20 to 35 s	Brake and drift	First hairpin: time slows at the entry until the player holds the pedal pad; release shows the mini-turbo
35 to 50 s	Special	Supply truck choice; the button pulses once when a target is in range
50 to 60 s	Everything together	A short mixed wave, then the director takes over

- Teach the downhill drop the first time it appears: a 10-second practice kicker with a prompt for one trick, then the real drop.
- Never show more than one instruction at a time, never more than 4 words while driving.

HUD for the new systems

- Pedal pad bottom right, special button above it, both mirrored by the handedness setting.
- Drift charge as sparks at the rear wheels (no meter needed), plus a thin ring under the car.
- A small speedometer next to the score, because speed is now the player's to manage.
- In drops: the Heat meter along the bottom edge and a trick name popup that never covers the landing zone.

Progression, economy and replay

Today a run leaves behind a best score, a separate daily best and a cash number that buys nothing. A 9 out of 10 game gives **every run, even a bad one, visible progress toward something the player wants.**

The meta loop

1. **Cash** from every run (10% of score, as now) plus mission rewards.
2. **Garage:** 5 cars at launch (Interceptor, Muscle, Rally, Supercar, Armored van), each with its own handling, mass and slots. The first unlock should land within 3 runs.
3. **Upgrades** per car: engine, armor, grip, drift, Slam power. 5 levels each, each level felt in play, not just a number.
4. **Loadout:** unlock weapons and gadgets for the six slots, and choose before each run. This is what makes runs different.
5. **Cosmetics:** paint, wraps, underglow, tire smoke color, exhaust flames, horns. Earned and optionally bought.
6. **Agent rank:** experience from every run unlocks new districts, drops and contracts.

Reasons to come back

Feature	How it works
Contracts	3 active spy missions ("Wreck the convoy before the dam", "Land 3 clean backflips in one drop", "Drift 2,000 m in Neon City") paying cash and rank
Daily Run	Same seed for everyone (already built), plus a Game Center leaderboard and a streak counter

Feature	How it works
Weekly event	Special rules: "drops only", "oil and flames only", "no guns"
Mastery	Each weapon and car levels up with use and unlocks a cosmetic
Hidden routes	Collectibles on the risky drop routes, tracked per district
Friend ghosts	Your friend's best distance marker appears on the road as you approach it
Highlight clip	The best 8 seconds of a run (biggest chain, best trick) saved as a shareable clip; <code>G.bestMoment</code> is already tracked but unused

Economy targets

- First car within 3 runs; a new unlock of some kind every run for the first hour; a full garage in about 15 to 20 hours.
- Never sell power that cannot be earned. Pick the business model before tuning prices: premium up front, or free with cosmetic purchases and one optional rewarded "continue" per run. No forced ads mid-run.

Performance and iOS

The graybox holds 60 fps in a browser, but the features asked for here (a chase camera, hills, tire smoke everywhere, a real landscape) need a 3D engine, and the iPhone needs native features the web cannot reach. **Choose the engine path now, because the art pipeline depends on it.**

Engine path

1. **Now:** port the graybox rules to three.js (WebGL) so the team can prototype corners, the chase camera and hills in the language it already uses. The road-space simulation carries over almost unchanged.
2. **At the fun gate, pick the shipping path** using the vertical slice on an iPhone 12:
 - If it holds 60 fps with effects on: ship the three.js game inside a native shell with native plugins for Core Haptics, Game Center, the audio session and gesture deferral.

- If it does not: port to Unity, which has mature car physics, particles and iOS tooling.

Fixes to make in the current code (small, do them in the next sprint)

Fix	Where	Effort
<code>#stage { flex-shrink: 0; }</code> so the game fills the screen undistorted	CSS	1 line
Contextual Slam with the corrected burst	<code>input.move</code> , <code>trySlam</code>	Small
Spawn fodder outside the player's lane; score wrecks by cause	<code>script</code> , <code>director</code> , <code>wreck</code>	Small
Draw the road as polygons, not 10-pixel slices	<code>drawRoad</code>	Medium
Interpolate rendering between physics steps	<code>simulate</code> , <code>render</code>	Small
Pools and in-place compaction instead of 21 <code>filter</code> copies per step	<code>physics</code>	Medium
Remember the last road knot index instead of scanning from the start	<code>Road.at</code>	Small
Run the gun inside the fixed physics step	<code>frame</code> , <code>gun</code>	Small
Shake inside the renderer with overscan, not CSS on the canvas	<code>render</code>	Small
Remove the "+0" pop; show the death replay or stop recording it	<code>contactPlayer</code> , <code>physics</code>	Small

iOS requirements for the shipping build

- Core Haptics patterns from the Game feel section, with the existing settings toggle.
- Defer system edge gestures and auto-hide the home indicator, so a thumb on the pedal pad never triggers the home swipe.
- Audio session set to ambient (respects the silent switch and the player's own music); pause on calls, Siri and backgrounding, then resume with the 3-2-1 countdown already built.
- Fill every screen from iPhone SE to Pro Max and iPad with no bars and no distortion; respect the Dynamic Island and safe areas.
- ProMotion: render at 120 Hz on Pro models, 60 Hz elsewhere, 30 Hz in Low Power Mode with reduced effects.

- Budgets on an iPhone 12: 60 fps for 99% of frames, at most 150 draw calls, 600 particles, 400 MB of memory, cold start to driving in 5 s, and 20 minutes of play without thermal throttling.
- Game Center leaderboards for the Daily Run and all-time best; fully playable offline.

Roadmap to 9 out of 10

Six sprints, each ending in a playtest gate on a real iPhone; a sprint is done when its gate passes, not when its tasks are checked off. Art runs in parallel from day one as direction and concepts, and only enters the game after the fun gate.

Art starts now as direction, and goes into the game only after the fun gate

GAMEPLAY

A. Fix and feel

Contextual Slam, player-caused kills, real threat
Full-screen fix, smooth road, interpolation

B. Driving model

Throttle, brake, heading steering, drift
Tire smoke, skid marks, mini-turbo, slipstream

C. 3D engine and corners

Port to three.js, road curvature, rotating camera
Corner types, technical sectors, hill crests

D. Downhill drop

Chase camera blend, kickers, air tricks
Heat meter, Overdrive, rival racers

E. Arsenal and enemies

Loadout slots, 20 new weapons, combos
10 new enemies, including the convoy boss

F. Meta and launch prep

Garage, upgrades, contracts, Game Center
Native shell or Unity port, TestFlight

ART

Art direction starts now

Mood board and 3 style frames, pick one
One-page art bible: palettes, roles, scale

Concept art

4 districts and 5 cars in the chosen style
Effects style: smoke, sparks, explosions

Readable blackout

Final car proportions and camera height
Simple roadside props for speed cues

Drop blackout

Alpine kicker lines, three routes
Sky, horizon and snow spray placeholders

Vertical slice

Neon City and one drop at final quality
Effects and audio at final quality

Production and polish

Remaining districts, cars, UI, store art
Lighting, detail levels, iPhone 12 perf pass

◆ Gate: idle driver under 5 wrecks a minute; under 1 accidental Slam per 10 minutes; lane dashes glide

◆ Gate: 4 of 5 testers drift without being told and say it feels like a car

◆ Gate: testers brake or drift at every hairpin; 60 fps on an iPhone 12

◆ Fun gate: 5 testers play 10 minutes and ask to keep going; the drop is what they talk about

◆ Gate: every tester finds a combo, and no two runs feel the same

◆ Launch gate: day-1 retention of 35% or more in TestFlight

roadmap · 6 sprints, 2 lanes, 6 gates

Sprint A is the cheapest and matters most: three of the five hard truths (the game plays itself, accidental Slams, the squashed screen) are small code changes.

The scorecard chart at the top holds today's score and the fun-gate target for every category; launch needs a 9 across the board.

Measure every playtest automatically (the bot logging used for this audit is a starting point): wrecks per minute with and without input, accidental Slams per minute, time with 2 or more cars on screen, longest quiet gap, armor lost per minute, cause of every death, and the second each new player quits their first run.